

Business Management

Lecture Notes

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Chapter 1

Lecture 1 – 07/09/2026

To manage a business, we first need to understand what a business is.

A business differs from economics: economics is the social science that studies how scarce resources are allocated, whereas a business is an organization that produces goods or services to satisfy customer needs and generate a profit.

Another perspective views a business as a legal entity. As an organization, it operates within a hierarchy that includes roles such as CEOs, managers, and employees.

1.0.1 Law

Law can be defined as:

- A set of rules designed to resolve conflicts of interest.
- An organizational framework to enforce those rules.
- A system of shared beliefs and social norms.

Generally, starting a business in a jurisdiction with strong legal protections is advantageous, as the law safeguards capital and intellectual property from theft or infringement.

For example, Artificial Intelligence (AI) regulation varies significantly by country. What is legal in one jurisdiction may be restricted or illegal in another, requiring founders to assess legal environments prior to launch.

1.0.2 Stakeholders

While shareholders hold equity shares and own part of the company, stakeholders include any individual or entity affected by the company's decisions and operations.

There are two primary types of stakeholders:

- **Internal Stakeholders:** Individuals within the organization, such as employees, managers, and owners.
- **External Stakeholders:** External entities affected by operations, including customers, suppliers, competitors, and regulatory bodies.

Each stakeholder group utilizes distinct legal and contractual mechanisms to protect their interests.

1.0.3 Financing the Company

- **Debt vs. Equity:** Debt refers to borrowed capital (e.g., from banks) that must be repaid with interest. Equity refers to capital raised from investors in exchange for ownership shares.
- **Equity Investors:** Investors provide capital in exchange for ownership shares, seeking long-term growth and accepting business risk. Key types include:
 - **Business Angels:** High-net-worth individuals who invest their personal capital in early-stage startups.
 - **Venture Capitalists (VCs):** Professional investment firms that deploy capital from institutional funds.
- **Debt Financing Instruments:** Different debt instruments carry distinct risk profiles and terms:
 - **Bonds:** Long-term debt securities issued to investors that pay periodic interest.
 - **Loans:** Direct debt agreements with fixed repayment schedules and interest rates.
 - **Lines of Credit:** Flexible credit arrangements allowing companies to draw funds up to a set limit as needed.

Chapter 2

Business Model

A business model is a conceptual framework describing how an organization creates, delivers, and captures value.

- **Revenue Assumptions:** What a company charges for (e.g., direct sales, subscription fees, data monetization, advertising space).
- **Cost Assumptions:** What a company incurs expenses for (e.g., raw materials, labor, marketing, R&D).

Business Model Definition

A business model defines who your target customer is, what value you create or add for them, and how you deliver that value at a sustainable cost.

The foundational step in business modeling is identifying customer needs. Sustainable profitability—rather than revenue alone—is essential, as profits generate the capital required for reinvestment and growth.

Example (Nespresso): Nespresso tested several business models that initially failed. They achieved success by repositioning coffee as a luxury experience—opening boutiques in premium retail locations worldwide and selling single-serve pods at a premium price point.

A robust business model also serves as a diagnostic tool to identify operational and legal risks.

“A business model describes how the business runs, whereas a competitive strategy defines how the business performs better than its competitors.”

2.0.1 Emerging Technologies

An emerging technology is a novel, rapidly growing technology characterized by fast development cycles and high market uncertainty.

Multiple technological solutions can address the same underlying customer need. For example, to commute to school, a person might buy a bicycle, rent one, walk, or hire a ride-sharing service. The optimal choice depends on constraints such as available budget and time.

2.0.2 Examples of Business Models

- **Crowdsourcing:** Outsourcing tasks or content creation to a distributed group of individuals, typically via online platforms (e.g., Wikipedia, YouTube).
- **Freemium:** Offering a basic version of a product or service for free while charging for premium features or higher usage tiers (e.g., Dropbox).
- **Subscription:** Charging customers a recurring fee (monthly or annually) for continuous access to a product or service (e.g., Netflix).
- **Negative Operating Cycle:** Receiving payment from customers before paying suppliers for goods, optimizing working capital (e.g., Amazon).
- **Disintermediation:** Removing intermediaries from the supply chain to sell directly to end consumers, reducing transaction costs (e.g., Dell).
- **Aggregator:** Consolidating information or service listings from multiple providers into a single platform for customer convenience (e.g., Airbnb).

Chapter 3

Lecture 2 – 09/09/2026 Business Model Canvas

The **Business Model Canvas (BMC)** is a visual tool used to describe, analyze, and develop a company's business model. It represents the main elements of a business and shows how they are connected.

The Business Model Canvas describes four main areas of a business:

- **Customers:** who the company serves;
- **Value:** what value the company provides to customers;
- **Infrastructure:** how the company creates and delivers value;
- **Finances:** how the company generates revenues and manages costs.

The Business Model Canvas helps entrepreneurs and managers visualize the business model, identify relationships between its different elements, and understand possible trade-offs.

3.0.1 Business Model vs. Business Plan

A **business model** describes how a company creates, delivers, and captures value. In particular, it explains how the company operates and how it generates profit.

A **business plan**, instead, is a formal document that describes the company's objectives, strategies, market analysis, operations, and expected financial performance.

Therefore:

- **Business model:** explains *how the business works*;
- **Business plan:** explains *how the business will be developed and managed*.

The Business Model Canvas is a tool that helps entrepreneurs and managers visualize and analyze the business model.

3.1 Business Model Examples

A company can generate revenues through different business models.

- **product:** The company sells a product or service directly to customers, generating

revenue through direct sales.

- **publicity / advertising:** The company generates revenue by selling advertising space or promoting products and services to a target audience.
- **fee / usage fee:** The company provides a service or platform and charges users a fee for access or for using the service. The customer pays according to the amount or frequency of usage.
- **subscription:** The company charges customers a recurring fee, usually monthly or annually, in exchange for continuous access to a product or service.
- **freemium:** The company offers a basic version of the product or service for free, while charging customers for additional or premium features. The objective is to attract a large number of users with the free version and convert some of them into paying customers.
- **accessories:** The company sells additional products or services that complement the main offering. Revenues are generated through **upselling** or **cross-selling** of accessories.

3.2 What is a Business Model?

The Business Model Canvas is composed of **nine building blocks**:

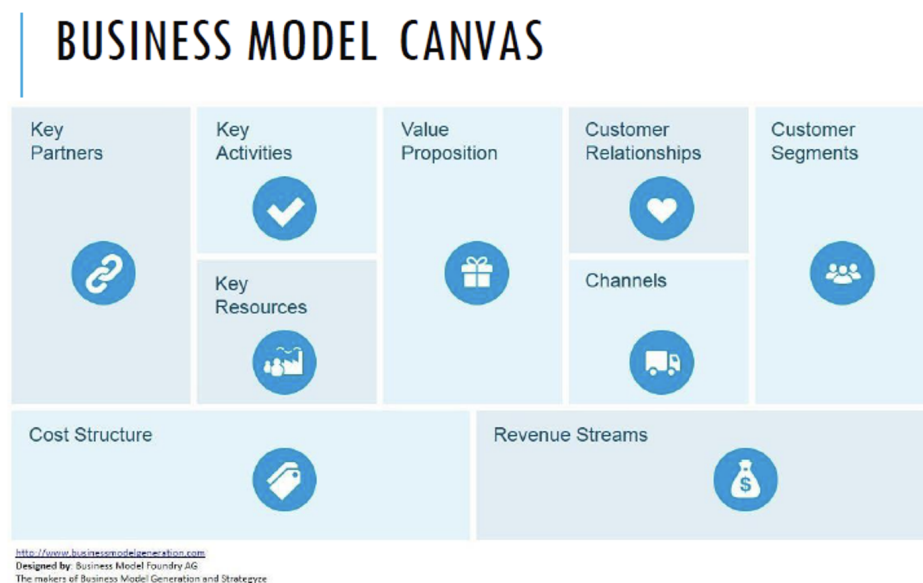


Figure 3.1: Business Model Canvas

These nine elements explain how a company creates, delivers, and captures value.

3.3 1. Value Proposition

The **value proposition** explains the value that a company offers to its customers and why customers should choose its product or service instead of competitors' alternatives.

The value proposition can be based on different factors, such as: price, quality, convenience, innovation, performance, customization, design, brand, accessibility, and risk

reduction.

A company can have **several value propositions** because different customer segments may have different needs and expectations.

The company should identify the value proposition that is the most profitable and sustainable for each target customer segment.

3.3.1 Types of Value Propositions

- **Novelty:** The product or service is new and innovative, offering a unique solution to a problem or need.
- **Performance:** The product or service offers superior performance or functionality compared with competitors.
- **Customization:** The product or service can be tailored to meet the specific needs or preferences of individual customers.
- **Design:** The product or service has an appealing aesthetic or user experience that differentiates it from competitors.
- **Brand / Status:** The product or service is associated with a prestigious brand or represents a status symbol that increases its perceived value.
- **Price:** The product or service is offered at a competitive or affordable price that appeals to cost-conscious customers.
- **Convenience:** The product or service is easy to access, use, or integrate into customers' lives, saving time or effort.
- **Risk Reduction:** The product or service reduces uncertainty or potential negative outcomes for customers,
- **Accessibility:** The product or service is made available to a wider range of customers, including customers with limited resources or special needs.
- **Cost Reduction:** The product or service helps customers save money or reduce expenses in the long term.

3.4 2. Customer Segments (For Whom?)

The **customer segment** is the group of people or organizations that a company aims to serve and for whom it creates value.

The market is divided into different groups of customers. These groups can have different needs, preferences, behaviors, and characteristics.

A segment can be identified according to:

- **Demographics:** age, gender, income, education, occupation, etc.;
- **Psychographics:** lifestyle, personality, values, interests, etc.;
- **Behavior:** purchasing habits, frequency of use, brand loyalty, etc.;
- **Needs:** specific problems or requirements that customers want to solve.

A **market segment** is a group of people who respond in a similar way to a marketing strategy.

For example, engineering students and architecture students may represent different customer segments because they can have different needs and preferences.

The customer segment can also be divided into **sub-segments** based on specific characteristics or behaviors.

In a market, there are different customer groups, and the company chooses a **target segment** on which to focus.

The **target segment** is the group of customers that the company specifically aims to serve and create value for.

3.4.1 Example: Ferrari

Ferrari focuses strongly on quality, performance, luxury, exclusivity, and brand/status.

Its target customers are generally high-income individuals who value: high performance, luxury, exclusivity, prestige, and status.

The target segment can also be divided into smaller sub-segments according to customers' specific preferences and motivations.

For Ferrari, customers may be particularly interested in the **status and prestige** associated with owning the brand.

3.5 3. Channels(How Do We Get It to Them?)

Channels describe how a company communicates with customers and how it delivers its value proposition to them.

Channels can be:

- direct or indirect;
- online or offline;
- owned by the company or shared with other companies.

Examples of channels include: physical stores, websites, mobile applications, social media, email marketing, distributors, partners, marketplaces, events, traditional advertising, and print media.

The company has to choose the channel that is the most **effective and efficient** for each customer segment.

3.5.1 Direct Channels

A **direct channel** is a channel through which the company sells or communicates directly with customers.

Examples include:

- a physical store owned by the company;
- the company's website;
- the company's mobile application;
- direct sales.

3.5.2 Indirect Channels

An **indirect channel** involves an intermediary between the company and the final customer.

Examples include: distributors, retailers, agents, business partners, and marketplaces.

3.5.3 Owned and Partner Channels

A company can use channels that it owns or channels shared with other companies.

For example:

- an owned website is an **owned channel**; legal forms
country and laws
company size
cost of complexity

3.5.4 Retail Store Example

A **retail store** is a business that sells products directly to customers.

The retailer can buy products from another company and resell them to customers, generating revenue through a **markup** or commission.

For example:

$$\text{Purchase Price} + \text{Markup} = \text{Selling Price}$$

The retail store can be: physical, online, or both.

A retailer can also provide additional services such as: customer support, returns, exchanges, warranties, marketing and promotional activities, and loyalty programs.

The channels and customer relationships should be adapted to the specific needs and characteristics of the target segment.

3.6 4. Customer Relationships

Customer relationships describe the type of relationship that a company establishes and maintains with its customers.

Customer relationships can be:

- personal or automated;
- transactional or long-term.

Examples include: personal assistance, dedicated personal assistance, self-service, automated services, loyalty programs, feedback systems, communities, and co-creation.

The company has to design the customer relationship that is most suitable for each customer segment.

For example, a cleaning-room worker and a university teacher may have very different needs and preferences. Therefore, the company may need to use different channels and customer relationships for different customer segments.

3.7 5. Revenue Streams

Revenue streams are the sources of income that a company generates from its business activities.

They explain **how the company makes money**.

Revenue streams can be based on different pricing models, such as: one-time sales, recurring subscriptions, licensing fees, advertising revenue, usage fees, and commissions.

Different customer segments or different value propositions can generate different revenue streams.

For example:

$$\text{Revenue} = \text{Price} \times \text{Quantity Sold}$$

However, a company may also generate recurring revenues through subscriptions or usage-based pricing.

3.8 6. Key Resources

Key resources are the assets that a company needs to create, deliver, and capture value.

Key resources can be classified into four main categories:

- **Physical resources;**
- **Intellectual resources;**
- **Human resources;**
- **Financial resources.**

3.8.1 Example: Electric Car Company

To produce electric cars, a company needs different types of key resources.

- **Physical:** factories, machinery, batteries, raw materials;
- **Intellectual:** patents, designs, software, technology;
- **Human:** engineers, designers, technicians;
- **Financial:** capital, loans, and investments.

Key resources can be owned by the company or accessed through partnerships and outsourcing.

3.9 7. Key Activities

Key activities are the most important actions that a company must perform to create and deliver its value proposition, reach customers, and generate revenues.

Examples of key activities include:

- production;
- research and development;

- marketing;
- sales;
- distribution;
- problem solving;
- platform management;
- customer support.

The key activities depend on the company's business model.

For example, for a manufacturing company, production may be one of the most important activities. For a technology company, software development and research may be more important.

3.10 8. Key Partnerships

Key partnerships are strategic alliances that a company forms with other organizations in order to improve its capabilities, reduce risks, access resources, or reach new markets.

Key partners can include: suppliers, distributors, technology providers, research institutions, joint-venture partners, and other companies.

These partnerships can help the company ensure a reliable supply chain, efficient charging infrastructure, and advanced vehicle features.

A company can start without key partnerships, but partnerships can become important when the company needs external resources, expertise, technology, distribution, or risk sharing.

3.10.1 Four Types of Partnerships

There are four main types of partnerships:

1. Strategic Alliances Between Non-Competitors

Two companies that operate in different industries or markets collaborate to achieve mutual benefits.

They are not direct competitors but can complement each other's activities.

2. Coopetition: Strategic Partnerships Between Competitors

Two companies that operate in the same industry or market collaborate to achieve common goals.

Examples include:

- joint research and development;
- shared marketing campaigns;
- co-branding;
- technology development.

This type of relationship can be described as **coopetition**, because companies cooperate while still competing in other areas.

3. Joint Ventures

Two or more companies create or participate in a separate business entity to pursue a specific project or business opportunity.

The partners share resources, risks, responsibilities, and potentially profits.

4. Buyer-Supplier Relationships / Bottleneck Partnerships

A company establishes a partnership with a supplier or another organization to secure access to an essential resource or overcome a specific constraint.

3.11 9. Cost Structure

The **cost structure** describes all the costs and expenses that a company incurs in order to operate its business model.

It includes the costs associated with: key resources, key activities, key partnerships, production, distribution, marketing, customer relationships, and administration. The cost structure can be divided into different types.

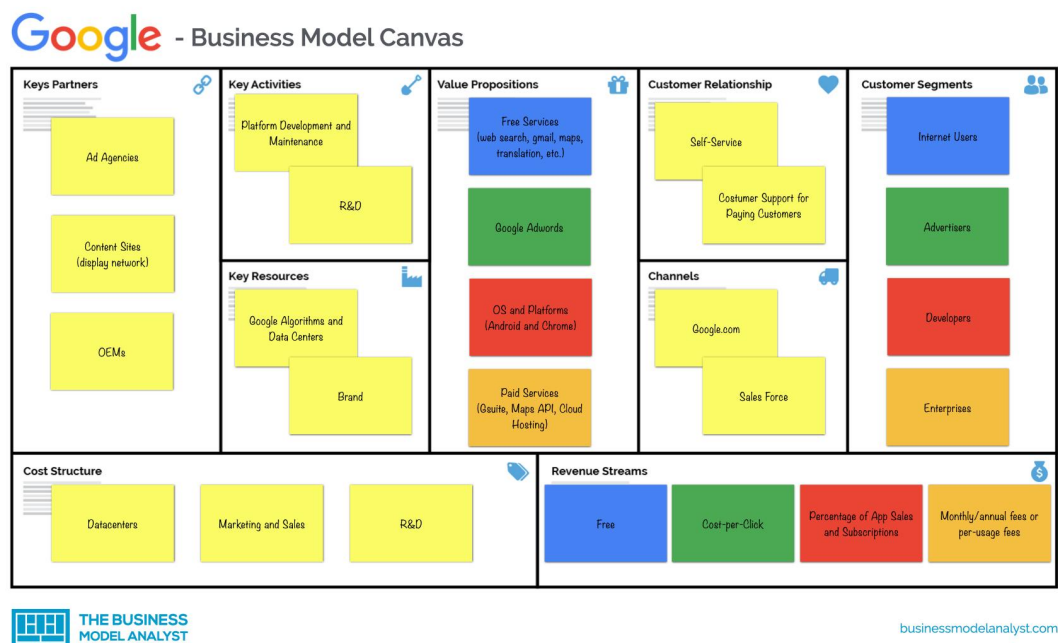


Figure 3.2: Cost Structure

Chapter 4

understand the business

Cue column	Note-taking area
goal of a business	the goal of a business is to create value for its customers and generate profit. profits is different with revenue. we can have revenue without profit.(paypal)
resources of a business	A business have 4 type of resources • money:allows the company to acquire other resources.
stakeholders of a business	• people: they are the ones who create value for the customers. • technology: allows the company to create value for the customers. • materials: they are the raw materials that are used to create value for the customers. These resources work together. No single resource is always “the most important”: its importance depends on the business and the situation stakeholders are divided in 4 type and depend on their goals. • shareholders: they are the owners of the company and they are interested about the profits. • executives: they are the CEO and they are interested about the growth of the company. • middle managers: they are the managers and they are interested about the performance of the company. • employees: they are the workers and they are interested about the stability of the company.

understand the business

Cue column

Note-taking area

flow of money

different types of companies

4.1 flow of money

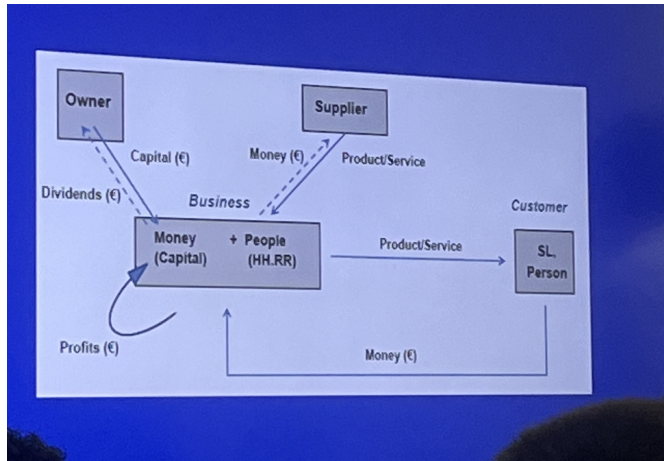


Figure 4.1: Cash may enter through customer payments, owner contributions or loans. It may leave to pay suppliers, employees, taxes, interest, lenders or investments. An owner contribution and a loan bring in cash, but are not sales revenue. Paying for equipment uses cash, but the equipment is normally recorded as an asset when purchased. Cash movements alone do not show whether the business made a profit.

Organisations have different aims:

- **For-profit companies** aim to create returns for their owners. They can reinvest profits or distribute them to owners.
- **Non-profit organisations** use any surplus to support their mission.
- **Cooperatives** are owned by their members and aim to benefit them.
- **Social enterprises** combine business activity with a social or environmental goal.
- **Government agencies**, such as public schools or hospitals, provide public services. Profit is not their main goal.

Summary

companies have different goals and different ways to use the profit.

understand a business

Cue column

Note-taking area

risk and profitability
over the time

4.2 timeline

legal forms

it is necessary to manage profitability and risk over the time. during the first years of a company, the risk is high and the profitability is low. after some years, the stability of company growth and for this reason the risk became low (TESLA). the profitability after some years can be reduced because the market is saturated and the competition is high.

country and laws

4.3 Legal forms

Working independently is not itself one universal legal form. A *freelancer* works independently for clients; the applicable registration and legal structure depend on the country. A *sole proprietor/sole trader* runs a business alone, generally without a separate company entity and with personal liability for business obligations. A *partnership* has two or more owners; liability depends on its particular form and local law. A *limited-liability company or corporation* is a separate entity whose owners generally have limited liability, subject to legal exceptions. It can have one or many owners: limited liability does **not** mean “owned by a small group”.

4.4 Country and rules

Company names and forms vary across legal systems. For example, *Inc.* is used in the US, *AG* in Germany and *S.p.A.* in Italy. These are examples of legal designations, not interchangeable structures. Registration, taxes, employment rules and reporting duties also depend on where the business operates.

Summary

Different legal forms determine how companies are owned and managed. Companies must follow the laws of the countries in which they operate, and larger organizations face higher costs of complexity because they manage more employees, products, and markets.

Chapter 5

understand the business - Finance

Cue column

Note-taking area

cost of complexity

Finance helps a company answer questions such as: How much money do we have? How much do we need? How can we obtain it?

5.1 Three useful financial statements

financial statements

Statement	Main question	Time frame
Balance sheet	What assets, liabilities and equity does the business have?	At a date
Income statement	What revenue and expenses produced profit or loss?	Over a period
Cash flow statement	How did cash change through operating, investing and financing activities?	Over a period

financial tools

These are a useful starting point, not necessarily the complete set of statements required by a reporting framework.

5.2 money values

- **Revenue:** income earned from selling goods or services.
- **Profit:** revenue minus expenses.
- **Cash:** money available now, for example in a bank account.
- **Debt:** money owed to creditors.
- **Equity:** the owners' interest in the company after liabilities are deducted from assets.

understand the business - Finance(2) -16/09/2026

Cue column

financial equation

difference between
bookkeeping and ac-
counting

Note-taking area

the financial equation

$$\boxed{\text{Assets} = \text{Liabilities} + \text{Equity.}}$$

this equation shows that the assets of a company are financed by the liabilities and the equity.

- the assets are the resources that a company owns
- the liabilities are the obligations that a company owes to others.
- the equity is the residual interest in the assets of a company after deducting liabilities.

bookkeeping

the bookkeeping is the process of recording financial transactions daily. Usually tools of this process are: daybook and general ledger. the daybook is a chronological record of all financial transactions. the general ledger is a record of all accounts and their balances. the chart of accounts is a list of all accounts used in the accounting system.

accounting

the accounting is the process of analyzing and interpreting transactions. It is used to prepare financial statements, such as the balance sheet, income statement, and cash flow statement.

five account families

- assets: resources owned by the company that have economic value and can be used to generate revenue (cash, machine...).
- liabilities: obligations that a company owes to others, such as loans, accounts payable, and taxes payable.
- equity: the residual interest in the assets of a company after deducting liabilities.
- revenue: income earned from selling goods or services.
- expenses: costs incurred in the process of generating revenue, such as salaries, rent, and utilities.

expense in the left and revenue in the right.

STATO PATRIMONIALE	
ATTIVITÀ	PASSIVITÀ
Immobilizzazioni	Patrimonio netto
Attività correnti (liquidità, rimanenze, crediti a breve termine e altre attività ricorrenti)	Debiti a medio-lungo termine
	Passività correnti (debiti a breve termine, debiti finanziari e altre passività correnti)
TOTALE ATTIVITÀ	TOTALE PASSIVITÀ

Figure 5.1: Account Families

Chapter 6

21/09/2026

balance Sheet is a financial statement that provides a snapshot of a company's financial position at a specific point in time. it records the current transactions (max one year) and is divided in two main sections: assets and liabilities/equity. the asset are written in order of liquidity, which means that the most liquid assets are listed first, followed by less liquid assets. This allows investors to quickly assess the liquidity of a company and its ability to meet short-term obligations. The order of liquidity is typically as follows: cash and cash equivalents, marketable securities, accounts receivable, inventory, and other current assets. By presenting assets in this order, investors can better understand the company's financial position and make informed investment decisions. the liabilities are also listed in order of due, which means that the liabilities that are due soonest are listed first, followed by those that are due later. This allows investors to quickly assess the company's ability to meet its short-term obligations and manage its long-term debt. The order of maturity is typically as follows: accounts payable, short-term debt, long-term debt, and other liabilities. By presenting liabilities in this order, investors can better understand the company's financial position and make informed investment decisions.

6.1 depreciation

the depreciation is the process of allocating the cost of a long-term asset over its useful life. It is a non-cash expense that reflects the decrease in value of an asset over time due to wear and tear, obsolescence, or other factors. Depreciation is recorded on the income statement as an expense, which reduces the company's net income. It is also recorded on the balance sheet as a reduction in the value of the asset. The purpose of depreciation is to match the cost of the asset with the revenue it generates over its useful life, providing a more accurate picture of the company's financial performance. the depreciation is considered an expense because it represents the allocation of the cost of a long-term asset over its useful life. When a company purchases a long-term asset, such as equipment or a building, it records the cost of the asset on its balance sheet as an asset. However, over time, the value of the asset decreases due to wear and tear, obsolescence, or other factors. To reflect this decrease in value, the company records depreciation as an expense on its income statement. This expense reduces the company's net income and reflects the true cost of using the asset over time.

every expense is a decrease of cash,

exercise: a company buys a machine for 3,000. the depreciation is linear the finale price

of the machine is 500 the time of use is 5 years the annual depreciation is $(3000-500)/5=$

6.2 the two clock of accounting

accounting has two clocks: the cash clock and the economic clock. The cash clock records transactions when cash is received or paid, while the economic clock records transactions when they are earned or incurred, regardless of when cash is exchanged. This means that revenue is recognized when it is earned, and expenses are recognized when they are incurred, even if payment has not yet been made. The accrual clock provides a more accurate picture of a company's financial performance and position, as it reflects the economic reality of transactions rather than just the timing of cash flows.

6.3 Purpose and main areas

Financial analysis examines a company's position and performance using financial statements and other information.

It helps managers, investors and creditors assess liquidity, solvency, profitability and efficiency. Analysts use ratios, changes over time and comparisons with similar businesses. A single ratio provides a clue, not a complete judgment.

- liquidity: ability to meet short-term obligations;
- solvency: ability to sustain operations and meet financial commitments over the longer term;
- profitability: relates income to sales or the resources used;
- efficiency: describes how well resources generate sales; asset turnover is one measure of efficiency

6.4 Liquidity: working capital and ratios

Cash is immediately available for payment, but other current assets may not be converted into cash quickly or at their recorded value. A profitable company can therefore experience a cash shortage.

Current assets include cash and items expected to be sold, consumed or realised in the normal operating cycle, or generally within twelve months. Current liabilities are obligations expected to be settled in that cycle or due in the short term. The twelve-month rule is a useful simplification, not the only classification criterion. Equity is not a current liability and is not subtracted when calculating working capital.

Net working capital is an amount expressed in currency:

$$\text{Working capital} = \text{Current assets} - \text{Current liabilities.}$$

A positive value means current assets exceed current liabilities at the balance-sheet date. It does not ensure that cash will be available when each bill becomes due: receivables may be collected late and inventory may sell slowly.

The current ratio compares the same categories as working capital but expresses their

relationship as a quotient:

$$\text{Current ratio} = \frac{\text{Current assets}}{\text{Current liabilities}}.$$

The quick ratio focuses on assets that can more readily be used to pay short-term obligations:

$$\text{Quick ratio} = \frac{\text{Cash} + \text{Short-term investments} + \text{Receivables}}{\text{Current liabilities}}.$$

It excludes inventory and other items, such as prepaid expenses, that cannot readily pay bills. An alternative simplified numerator is current assets minus inventory and prepayments, when those categories explain the difference. A ratio above 1 says that the assets included in its numerator exceed current liabilities on that date; it does not guarantee timely payment. The quality of receivables, payment dates, cash flows and industry norms also matter. Neither ratio is defined in the usual way when current liabilities are zero.

6.5 Solvency and indebtedness

Solvency depends on obligations, the value and composition of assets, and the capacity to generate future cash. Merely having total assets above long-term debt is not a sufficient assessment. Borrowing can finance growth but creates payment commitments that increase risk if income or cash flows fall.

The term “financial leverage ratio” is ambiguous: it can refer to different measures. If the numerator includes *all liabilities*, the first ratio should be labelled liabilities-to-equity:

$$\text{Liabilities-to-equity} = \frac{\text{Total liabilities}}{\text{Total equity}}.$$

A value of 0.8 means that the company has 80 cents of liabilities for each euro of book equity. Some analysts use only interest-bearing borrowing in the numerator and call the result debt-to-equity; this is a different definition. The equity multiplier, assets divided by equity, is another leverage measure. A debt-to-equity figure is difficult to interpret in the usual way when equity is zero or negative.

If “debt” is defined as all liabilities, the corresponding ratio to assets is

$$\text{Liabilities-to-assets} = \frac{\text{Total liabilities}}{\text{Total assets}}.$$

This indicates the share of book assets financed through obligations to others. If a course uses *debt ratio* for interest-bearing borrowing divided by assets, that narrower definition should be stated and used consistently. A higher ratio can indicate greater financial risk, but its meaning depends on the business model, cash-flow stability and comparison group.

6.6 Operating efficiency: asset turnover

Total asset turnover measures sales generated per unit of assets during a period:

$$\begin{aligned} \text{Total asset turnover} &= \frac{\text{Net sales}}{\text{Average total assets}}, \\ \text{Average total assets} &= \frac{\text{Opening assets} + \text{Closing assets}}{2}. \end{aligned}$$

A high turnover can reflect effective use of assets, but it does not show the profit margin earned on those sales. Because net sales cover a period and the balance sheet measures assets at points in time, the average of opening and closing assets is commonly used.

6.7 Earnings: EBIT and EBT

EBIT means *earnings before interest and taxes*. It is an amount, not a ratio, and is commonly used to examine performance before financing cost and income tax. EBT means *earnings before taxes*: it reflects interest expense and other relevant items before income tax. In a simplified income statement, EBIT minus net interest expense gives EBT, and EBT minus income tax gives net income. Actual statements can contain other non-operating items, so this sequence needs to be checked against the statement being analysed.

6.8 Returns: ROI, ROA and ROE

For a particular project, one simple definition of return on investment is

$$\text{ROI} = \frac{\text{Net gain from the investment}}{\text{Cost of the investment}} \times 100\%.$$

Net gain means proceeds or benefits minus the investment cost and any other costs included in the chosen definition. If proceeds are 120 and the investment costs 100, net gain is 20 and ROI is 20%. Managerial accounting may instead use “ROI” for a segment’s operating income divided by its average invested assets. Its numerator, denominator and period must therefore be specified. A simple ROI also ignores when cash flows occur.

Using net income as the measure of profit, return on assets and return on equity can be written as

$$\begin{aligned}\text{ROA} &= \frac{\text{Net income}}{\text{Average total assets}} \times 100\%, \\ \text{ROE} &= \frac{\text{Net income}}{\text{Average equity}} \times 100\%.\end{aligned}$$

The averages ordinarily use opening and closing balances. ROA measures net income relative to the assets employed; ROE measures net income relative to owners’ equity. Some courses use an operating-profit version of ROA: the two versions should not be mixed in a comparison. Debt can amplify ROE and also amplify losses, so a high ROE alone does not establish that a business is stronger. ROE becomes hard to interpret when equity is very small or negative.

6.9 Interpreting comparisons

EBIT and EBT are absolute earnings amounts, while ROI, ROA and ROE are ratios. Ratios can help compare businesses of different sizes, and EBIT can also be compared across businesses when its definition and the underlying accounting are aligned. All these measures can be examined over time. Useful comparisons require compatible definitions, the same period and, generally, businesses in similar industries. No metric is automatically more useful than the others in every situation.